

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 9: MATRICES (2) – EIGENVALUES, EIGENVECTORS, CHARACTERISTICS EQUATION

1. The matrix P is given by

$$P = \begin{pmatrix} 1 & 4 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

- (a) State the eigenvalues of P . ($\mathbf{1}, -\mathbf{1}, \mathbf{2}$)

[1]

- (b)

[4]

Use the characteristics equation of P to find P^{-1} . ($P^{-1} = \begin{pmatrix} \mathbf{1} & \mathbf{4} & \mathbf{-3} \\ \mathbf{0} & \mathbf{-1} & \mathbf{\frac{1}{2}} \\ \mathbf{0} & \mathbf{0} & \mathbf{\frac{1}{2}} \end{pmatrix}$)

The 3×3 matrix A has distinct eigenvalues $b, -1, 1$ with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix},$$

respectively.

- (c) Find A in terms of b . $\begin{pmatrix} \mathbf{b} & \mathbf{4b+4} & \mathbf{-3b-1} \\ \mathbf{0} & \mathbf{-1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$

[4]

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2. The matrix A is given by

$$A = \begin{pmatrix} a & 2a+5 & a+1 \\ 0 & -4 & 0 \\ 0 & 3 & -1 \end{pmatrix}.$$

- (b) Show that the eigenvalues of A are $a, -1$ and -4 .

[2]

- (c) Find a matrix P such that

$$A = P \begin{pmatrix} a & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -4 \end{pmatrix} P^{-1}.$$

[5]

- (d)

Use the characteristic equation of A to find A^{-1} . ($A^{-1} = \frac{1}{4a} \begin{pmatrix} \mathbf{4} & \mathbf{5a+8} & \mathbf{4a+4} \\ \mathbf{0} & \mathbf{-a} & \mathbf{0} \\ \mathbf{0} & \mathbf{-3a} & \mathbf{-4a} \end{pmatrix}$)

[6]

$$P = \begin{pmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{-1} \\ \mathbf{0} & \mathbf{-1} & \mathbf{1} \end{pmatrix}$$

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3. The matrix A is given by

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 6 & -1 \\ 0 & 0 & -2 \end{pmatrix}.$$

- (b) Use the characteristics equation of A to find the inverse of A^2 . $\left((A^{-1})^2 = \right.$ [4]

$$\frac{1}{36} \begin{pmatrix} 4 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 9 \end{pmatrix} \Bigg)$$

- (c) Find a matrix P and a diagonal matrix D such that $A^5 = PDP^{-1}$. $\left(P = \right.$ [7]

$$\begin{pmatrix} 1 & 1 & -9 \\ 0 & 3 & 5 \\ 0 & 0 & 40 \end{pmatrix}, D = \begin{pmatrix} 243 & 0 & 0 \\ 0 & 7776 & 0 \\ 0 & 0 & -32 \end{pmatrix} \Bigg)$$

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4. The matrix A is given by

$$A = \begin{pmatrix} 5 & -1 & 7 \\ 0 & 6 & 0 \\ 7 & 7 & 5 \end{pmatrix}.$$

- (a) Find the eigenvalues of A . $(-2, 6, 12)$ [4]

- (b) Use the characteristics equation of A to find A^{-1} . $\left(A^{-1} = \frac{1}{24} \begin{pmatrix} -5 & -9 & 7 \\ 0 & 4 & 0 \\ 7 & 7 & -5 \end{pmatrix} \right)$ [4]

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5. (b) The matrix A is given by

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}.$$

- (i) Find the eigenvalues of A . $(1, 2, 3)$ [4]

- (ii) Use the characteristics equation of A to find A^{-1} . $\left(A^{-1} = \frac{1}{6} \begin{pmatrix} 4 & -1 & -2 \\ -2 & 5 & -2 \\ 2 & -2 & 2 \end{pmatrix} \right)$ [4]

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6. The matrix M is defined by

$$M = \begin{pmatrix} 2 & m & 1 \\ 0 & m & 7 \\ 0 & 0 & 1 \end{pmatrix},$$

where $m \neq 0, 1, 2$.

- (i) Find a matrix P and a diagonal matrix D such that $M = PDP^{-1}$. $\left(P = \right.$ [7]

$$\begin{pmatrix} 1 & m & 6m+1 \\ 0 & m-2 & -7 \\ 0 & 0 & m-1 \end{pmatrix}, D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- (ii) Find M^7P . [3]

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7. Consider the system of equations

$$\begin{aligned} x + 5y + z &= -1, \\ x - 2y - 2z &= 0, \\ 2x + 3y + \theta z &= \theta. \end{aligned}$$

- (ii) Solve the system of equations when $\theta \neq -1$. $\left(z = 1, y = -\frac{4}{7}, x = \frac{6}{7} \right)$ [3]

- (iii) Find the general solution when $\theta = -1$. $\left(z = t, y = -\frac{3t+1}{7}, x = \frac{8t-2}{7} \right)$ [3]

- (iv) Show that if $\theta = -1$ and $\phi \neq -1$ then $Ax = \begin{pmatrix} -1 \\ 0 \\ \phi \end{pmatrix}$ has no solution. [2]

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8. It is given that e is an eigenvector of the matrix A , with corresponding eigenvalue λ .

- (i) Show that e is an eigenvector of A^2 , with corresponding eigenvalue λ^2 . [2]

The matrices A and B are given by

$$A = \begin{pmatrix} n & 1 & 3 \\ 0 & 2n & 0 \\ 0 & 0 & 3n \end{pmatrix} \quad \text{and} \quad B = (A + nI)^2.$$

where I is the 3×3 identity matrix and n is a non-zero integer.

- (ii) Find, in terms of n , a non-singular matrix P and a diagonal matrix D such that $B = PDP^{-1}$. [8]

$$PDP^{-1}. \left(P = \begin{pmatrix} 1 & 1 & 3 \\ 0 & n & 0 \\ 0 & 0 & 2n \end{pmatrix}, D = \begin{pmatrix} (2n)^2 & 0 & 0 \\ 0 & (3n)^2 & 0 \\ 0 & 0 & (4n)^2 \end{pmatrix} \right)$$

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9. A 3×3 matrix A has distinct eigenvalues 2, 1, 3, with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ b \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

respectively, where b is a positive constant.

- (i) Find A in terms of b . $\left(A = \frac{1}{b+1} \begin{pmatrix} 1+2b & 1 & 1 \\ -b & 3b+2 & -1 \\ 2b & -2b & 3+b \end{pmatrix} \right)$ [9]

(ii) Find $A^{-1} \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} \cdot \left(\frac{2}{3} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right)$ [2]

(iii) It is given that

$$A^n \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} \quad \text{and} \quad A^n \begin{pmatrix} -1 \\ 0 \\ b \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ b^{-1} \end{pmatrix}.$$

[3]

Find the values of n and b . ($n = 2, b = 1$)

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10. It is given that λ is an eigenvalue of the matrix A with e as a corresponding eigenvector, and μ is an eigenvalue of the matrix B for which e is also a corresponding eigenvector.

(i) Show that $\lambda + \mu$ is an eigenvalue of the matrix $A + B$ with e as a corresponding eigenvector. [2]

The matrix A , given by

$$A = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 2 & 3 \\ 1 & 0 & 2 \end{pmatrix}$$

has $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ as eigenvectors.

(ii) Find the corresponding eigenvalues. (**3, 1, 2**) [3]

The matrix B has eigenvalues 4, 5 and 1 with corresponding eigenvectors $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ respectively.

(iii) Find a matrix P and a diagonal matrix D such that $(A + B)^3 = PDP^{-1}$. ($P =$ [3]

$$\begin{pmatrix} 1 & 1 & 0 \\ 2 & 4 & 1 \\ 1 & -1 & 0 \end{pmatrix}, D = \begin{pmatrix} 343 & 0 & 0 \\ 0 & 216 & 0 \\ 0 & 0 & 27 \end{pmatrix}$$

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11. It is given that

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(i) Find the eigenvalue of A corresponding to the eigenvector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. (**2**) [1]

- (ii) Write down the negative eigenvalue of A and find a corresponding eigenvector. [3]

$$\left(-2, \begin{pmatrix} 3 \\ -4 \\ 0 \end{pmatrix}\right)$$

- (iii) Find an eigenvalue and a corresponding eigenvector of the matrix $A + A^6$. [2]

$$\left(66, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ OR } 62, \begin{pmatrix} 3 \\ -4 \\ 0 \end{pmatrix} \text{ OR } 2, \begin{pmatrix} -6 \\ 1 \\ 3 \end{pmatrix}\right)$$

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12. It is given that e is an eigenvector of the matrix A , with corresponding eigenvalue of λ .

- (i) Write down another eigenvector of A corresponding to λ . [1]
(ke , where k is any scalar)

- (ii) Write down an eigenvector and corresponding eigenvalue of A^n , where n is a positive integer. (e, λ^n) [2]

$$\text{Let } A = \begin{pmatrix} 3 & 0 & 0 \\ 2 & 7 & 0 \\ 4 & 8 & 1 \end{pmatrix}.$$

- (iii) Find a matrix P and a diagonal matrix D such that $A^n = PDP^{-1}$. [7]

$$\left(P = \begin{pmatrix} -2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 4 & 1 \end{pmatrix}, D = \begin{pmatrix} 3^n & 0 & 0 \\ 0 & 7^n & 0 \\ 0 & 0 & 1 \end{pmatrix}\right)$$

- (iv) Determine the set of values of the real constant k such that

$$\sum_{n=1}^{\infty} k^n (A^n - kA^{n+1}) = kA. \quad [4]$$

$$\left(-\frac{1}{7} < k < \frac{1}{7}\right)$$

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13. It is given that e is an eigenvector of the matrix A with corresponding eigenvalue of λ .

- (i) Show that e is an eigenvector of A^3 and state the corresponding eigenvalue. (λ^3) [3]

It is given that

$$A = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}.$$

- (ii) Find a matrix P and a diagonal matrix D such that

$$A^3 + I = PDP^{-1},$$

$$\text{where } I \text{ is the } 2 \times 2 \text{ identity matrix. } \left(2, 3; P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 9 & 0 \\ 0 & 28 \end{pmatrix}\right) \quad [5]$$

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14. (i) The vector e is an eigenvector of the matrix A , with corresponding eigenvalue λ , and is

also an eigenvector of the matrix B , with corresponding eigenvalue of μ . Show that e is an eigenvector of the matrix AB with corresponding eigenvalue $\lambda\mu$. [3]

- (ii) Find the eigenvalues and corresponding eigenvectors of the matrix A , where

$$A = \begin{pmatrix} 0 & 1 & 3 \\ 3 & 2 & -3 \\ 1 & 1 & 2 \end{pmatrix}. \quad [6]$$

$$\left(-1, 2, 3; \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)$$

- (iii) The matrix B , where

$$B = \begin{pmatrix} 3 & 6 & 1 \\ 1 & -2 & -1 \\ 6 & 6 & -2 \end{pmatrix}.$$

has eigenvectors $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Find the eigenvalues of the matrix AB , and state corresponding eigenvectors. [4]

$$\left(3, -4, 12; \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)$$

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15. The matrix A , given by

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 6 & 4 & -6 \\ 6 & 5 & -7 \end{pmatrix},$$

has eigenvalues 1, -1 and -2 .

- (i) Find a set of corresponding eigenvectors. $\left(\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right)$ [4]

- (ii) The matrix B is given by $B = A - 2I$, where I is the 3×3 identity matrix. Write down the eigenvalues of B , and state a set of corresponding eigenvectors. [2]

$$\left(-1, -3, -4; \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right)$$

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16. The matrix A is given by

$$A = \begin{pmatrix} 6 & -8 & 7 \\ 7 & -9 & 7 \\ 6 & -6 & 5 \end{pmatrix}.$$

(i) Given that $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector of A , find the corresponding eigenvalue. (-2) [2]

(ii) Given also that -1 is an eigenvalue of A , find a corresponding eigenvector. $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ [2]

(iii) It is given that the determinant of A is equal to product of the eigenvalues of A . Use this result to find the third eigenvalue of A , and find also a corresponding eigenvector. [3]

$$\begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$$

(iv) Write down matrices P and D such that $P^{-1}AP = D$, where D is a diagonal matrix, and hence find the matrix A^n in terms of n , where n is a positive integer. [6]

$$\left(\begin{array}{l} P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix}, D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \\ A^n = \begin{pmatrix} 5^n + (-1)^n - (-2)^n & 2 \cdot (-2)^n + (-1)^{n+1} - 5^n & 5^n - (-2)^n \\ 5^n - (-2)^n & 2 \cdot (-2)^n - 5^n & 5^n - (-2)^n \\ 5^n - (-1)^n & (-1)^n - 5^n & 5^n \end{pmatrix} \end{array} \right)$$

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17. Find a matrix A whose eigenvalues are $-1, 1, 2$ and for which corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

respectively.

$$\left(A = \begin{pmatrix} -1 & 2 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \right)$$

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18. Write down the eigenvalues of the matrix A , where

$$A = \begin{pmatrix} -2 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix},$$

and find corresponding eigenvectors.

Find a matrix P and a diagonal matrix D such that $P^{-1}AP = D$, and hence find the matrix A^n , where n is a positive integer.

$$\left(\begin{array}{l} -2, -1, 1, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}; P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ A^{2n} = \begin{pmatrix} (-2)^n & -(-2)^n + (-1)^n & (-2)^n + (-1)^n \\ 0 & (-1)^n & (-1)^{n+1} + 1 \\ 0 & 0 & 1 \end{pmatrix} \end{array} \right)$$

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19. The matrix A , where

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 10 & -7 & 10 \\ 7 & -5 & 8 \end{pmatrix},$$

has eigenvalues 1 and 3. Find corresponding eigenvectors. $\left(\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right)$ [3]

It is given that $\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ is an eigenvector of A . Find the corresponding eigenvalue. ($\lambda = -2$) [2]

Find a diagonal matrix D and matrices P and P^{-1} such that $P^{-1}AP = D$. [5]

$$\left(\begin{array}{l} P = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}, D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \\ P^{-1} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 0 \\ 2 & -1 & 2 \end{pmatrix} \end{array} \right)$$

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20. The matrix A is given by

$$A = \begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix}.$$

The matrix A has an eigenvector $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$. Find the corresponding eigenvalue. ($\lambda = -3$) [2]

The matrix A also has eigenvalues 4 and 6. Find corresponding eigenvectors. $\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right)$ [3]

Hence find a matrix P such that $A = PDP^{-1}$, where D is a diagonal matrix which is to be determined. [2]

$$\left(P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & 0 & -2 \end{pmatrix}, D = \begin{pmatrix} -3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{pmatrix} \right)$$

The matrix B is such that $B = QAQ^{-1}$, where

$$Q = \begin{pmatrix} 4 & 11 & 5 \\ 1 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}.$$

By using the expression PDP^{-1} for A , find the set of eigenvalues and a corresponding set of eigenvectors for B . [5]

$$\left(-3, 4, 6; \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 15 \\ 5 \\ 3 \end{pmatrix}, \begin{pmatrix} 17 \\ 7 \\ 3 \end{pmatrix} \right)$$

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21. One of the eigenvalues of the matrix M , where

$$M = \begin{pmatrix} 3 & -4 & 2 \\ -4 & \alpha & 6 \\ 2 & 6 & -2 \end{pmatrix},$$

is -9 . Find the value of α . ($\alpha = -1$) [3]

Find

(i) the other two eigenvalues, λ_1 and λ_2 , of M , where $\lambda_1 > \lambda_2$, ($\lambda_1 = 6, \lambda_2 = 3$) [5]

(ii) corresponding eigenvectors for all three eigenvalues of M . $\left(\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \right)$ [3]

It is given that $x = ae_1 + be_2$, where e_1 and e_2 are eigenvectors of M corresponding to the eigenvalues λ_1 and λ_2 respectively, and a and b are scalar constants. Show that $Mx = pe_1 + qe_2$, expressing p and q in terms of a and b . ($p = 6a, q = 3b$) [3]

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22. The square matrix A has λ as an eigenvalue with e as a corresponding eigenvector. Show that if A is non-singular then

(i) $\lambda \neq 0$, [2]

(ii) the matrix A^{-1} has λ^{-1} as an eigenvalue with e as a corresponding eigenvector. [2]

The 3×3 matrices A and B are given by

$$A = \begin{pmatrix} -2 & 2 & -4 \\ 0 & -1 & 5 \\ 0 & 0 & 3 \end{pmatrix} \quad \text{and} \quad B = (A + 3I)^{-1},$$

where I is the 3×3 identity matrix. Find a non-singular matrix P , and a diagonal matrix D , such that $B = PDP^{-1}$. [10]

$$\left(P = \begin{pmatrix} 1 & 2 & -6 \\ 0 & 1 & 25 \\ 0 & 0 & 20 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/6 \end{pmatrix} \right)$$

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23. The matrix M , where

$$M = \begin{pmatrix} -2 & 2 & 2 \\ 2 & 1 & 2 \\ -3 & -6 & -7 \end{pmatrix},$$

has an eigenvector $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. Find the corresponding eigenvalue.

It is given that if the eigenvalues of a general 3×3 matrix A , where

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

are λ_1, λ_2 and λ_3 then

$$\lambda_1 + \lambda_2 + \lambda_3 = a + e + i$$

and

the determinant of A has the value $\lambda_1 \lambda_2 \lambda_3$.

Use these results to find the other two eigenvalues of the matrix M , and find corresponding

eigenvectors. $\left(-1; -3; -4; \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right)$

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- 24.** It is given that λ is an eigenvalue of the non-singular square matrix A , with corresponding eigenvector e . Show that λ^{-1} is an eigenvalue of A^{-1} for which e is a corresponding eigenvector.

Deduce that $\lambda + \lambda^{-1}$ is an eigenvalue of $A + A^{-1}$.

It is given that 1 is an eigenvalue of the matrix A , where

$$A = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 2 & 3 \\ 1 & 0 & 2 \end{pmatrix}.$$

Find a corresponding eigenvector. $\begin{pmatrix} -1 \\ -4 \\ 1 \end{pmatrix}$

It is also given that $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix A . Find the corresponding eigenvalues. **(2, 3)**

Hence find a matrix P and a diagonal matrix D such that

$$(A + A^{-1})^3 = PDP^{-1}.$$

$$\left(P = \begin{pmatrix} -1 & 0 & 1 \\ -4 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}, D = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 125/8 & 0 \\ 0 & 0 & 1000/27 \end{pmatrix} \right)$$

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25. The square matrix A has λ as an eigenvalue with e as a corresponding eigenvector. Show that e is an eigenvector of A^2 and state the corresponding eigenvalue.

Find the eigenvalues of the matrix B , where

$$B = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}.$$

Find the eigenvalues of $B^4 + 2B^2 + 3I$, where I is the 3×3 identity matrix.
($\lambda^2; -2, 1, 4; 6, 27, 291$)

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26. The vector e is an eigenvector of each of the 3×3 matrices A and B , with corresponding eigenvalues λ and μ respectively. Justifying your answer, state an eigenvalue of $A + B$. [3]

The matrix A , where

$$A = \begin{pmatrix} 6 & -1 & -6 \\ 1 & 0 & -2 \\ 3 & -1 & -3 \end{pmatrix},$$

has eigenvectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$. Find the corresponding eigenvalues. [4]

The matrix B , where

$$B = \begin{pmatrix} 8 & -2 & -8 \\ 2 & 0 & -4 \\ 4 & -2 & -4 \end{pmatrix},$$

also has eigenvectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$, which $-2, 2, 4$ respectively, are corresponding eigenvalues. The matrix M is given by $M = A + B - 5I$, where I is the 3×3 identity matrix. State the eigenvalues of M . [1]

Find matrices R and S and a diagonal matrix D such that $M^5 = RDS$. [6]

[You should show clearly all the elements of the matrices R, S and D .]

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27. The matrix A is given by

$$A = \begin{pmatrix} 4 & -5 & 3 \\ 3 & -4 & 3 \\ 1 & -1 & 2 \end{pmatrix}.$$

Show that $e = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is an eigenvector of A and state the corresponding eigenvalue. ($\lambda = 2$) [2]

Find the other two eigenvalues of A . ($1, -1$) [4]

The matrix B is given by

$$B = \begin{pmatrix} -1 & 4 & 0 \\ -1 & 3 & 1 \\ 1 & -1 & 3 \end{pmatrix}.$$

Show that e is an eigenvector of B and deduce an eigenvector of the matrix AB , stating the corresponding eigenvalue. $\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, -6 \right)$ [3]

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28. The square matrix A has an eigenvalue λ with corresponding eigenvector e . The non-singular matrix M is of the same order as A . Show that Me is an eigenvector of the matrix B , where $B = MAM^{-1}$, and the λ is the corresponding eigenvalue. [3]

Let

$$A = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

6Write down the eigenvalues of A and obtain corresponding eigenvectors. $\left(\lambda = -1, 1, 2; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \right)$ [4]

Given that

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

find the eigenvalues and corresponding eigenvectors of B . $\left(\lambda = -1, 1, 2; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix} \right)$

[4]
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29. Write down the eigenvalues of the matrix A , where

$$A = \begin{pmatrix} 1 & 4 & -16 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{pmatrix}. \quad [1]$$

Find corresponding eigenvectors. [4]

$$\left(\lambda = 1, 2, 3; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} \right)$$

Let n be a positive integer. Write down a matrix P and a diagonal matrix D such that

$$A^n = PDP^{-1}, \quad [2]$$

$$\left(P = \begin{pmatrix} 1 & 4 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2^n & 0 \\ 0 & 0 & 3^n \end{pmatrix} \right)$$

Find P^{-1} and A^n . [5]

$$\left(P^{-1} = \begin{pmatrix} 1 & -4 & 14 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}, A^n = \begin{pmatrix} 1 & -4 + 4 \cdot 2^n & 14 - 12 \cdot 2^n - 2 \cdot 3^n \\ 0 & 2^n & -3 \cdot 2^n + 3^{n+1} \\ 0 & 0 & 3^n \end{pmatrix} \right)$$

Hence find $\lim_{n \rightarrow \infty} (3^{-n} A^n)$. [1]

$$\begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

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30. The vector e is an eigenvector of each of the vector $n \times n$ matrices A and B , with corresponding eigenvalues λ and μ respectively. Prove that e is an eigenvector of the matrix AB with eigenvalue $\lambda\mu$. [2]

It is given that the matrix A , where

$$A = \begin{pmatrix} 3 & 2 & 2 \\ -2 & -2 & -2 \\ 1 & 2 & 2 \end{pmatrix},$$

has eigenvectors $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. Find the corresponding eigenvalues. ($\lambda = 0, 1$) [2]

Given that 2 is also an eigenvalue of A , find a corresponding eigenvector. $\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ [2]

The matrix B , where

$$B = \begin{pmatrix} -1 & 2 & 2 \\ 2 & 2 & 2 \\ -3 & -6 & -6 \end{pmatrix},$$

has the same eigenvectors as A . Given that $AB = C$, find a non-singular matrix P and a diagonal matrix D such that

$$P^{-1}C^2P = D. \quad [8]$$

$$\left(P = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}, D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{pmatrix} \right)$$

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31. A matrix A has eigenvalues $-1, 1$ and 2 . With corresponding eigenvectors.

$$\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix},$$

Respectively. Find A .

$$A = \begin{pmatrix} 3 & 4 & 2 \\ -11 & -27 & -13 \\ 21 & 54 & 26 \end{pmatrix}$$

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32. The matrix A has an eigenvalue λ with corresponding eigenvector e . Prove that the matrix $(A + kI)$, where k is a real constant and I is the identity matrix, has an eigenvalue $(\lambda + k)$ with corresponding eigenvector e .

The matrix B is given by

$$B = \begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix}.$$

Two of the eigenvalues of B are -3 and 4 . Find corresponding eigenvectors. $\left(\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right)$

Given that $\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$ is an eigenvector of B , find the corresponding eigenvalue. ($\lambda = 6$)

Hence find the eigenvalues of C , where

$$C = \begin{pmatrix} -1 & 2 & -3 \\ 2 & -1 & 3 \\ -3 & 3 & 0 \end{pmatrix},$$

and state corresponding eigenvectors.

$$\left(-6, 1, 3; \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right)$$

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33. The vector e is an eigenvector of the matrix A , with corresponding eigenvalue λ , and is also an eigenvector of the matrix B , with corresponding eigenvalue μ . Show that e is an eigenvector of the matrix AB with corresponding eigenvalue $\lambda\mu$. [2]

State the eigenvalues of the matrix C , where

$$C = \begin{pmatrix} -1 & -1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix},$$

[4]

and find corresponding eigenvectors.

$$\left(-1, 1, 2; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 6 \\ 3 \end{pmatrix} \right)$$

Show that $\begin{pmatrix} 1 \\ 6 \\ 3 \end{pmatrix}$ is an eigenvector of the matrix D , where

$$D = \begin{pmatrix} 1 & -1 & 1 \\ -6 & -3 & 4 \\ -9 & -3 & 7 \end{pmatrix}.$$

and state the corresponding eigenvalue. (-2) [3]

Hence state an eigenvector of the matrix CD and give the corresponding eigenvalue. [2]

$$\left(\begin{pmatrix} 1 \\ 6 \\ 3 \end{pmatrix}, -4 \right)$$

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34. Find the eigenvalues and corresponding eigenvectors of the matrix A , where

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}. \quad [7]$$

$$\left(\lambda = 1, 2, 3; \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right)$$

The linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by $x \mapsto Ax$. Let e, f be two linearly independent eigenvectors of A , with corresponding eigenvalues λ and μ respectively, and let

Π be the plane, through the origin, containing e and f . By considering the parametric equation of Π , show that all points of Π are mapped by T onto points of Π . [3]

Find cartesian equations of three planes, each with the property that all points of the plane are mapped by T onto points of the same plane. ($x - y - 2z = 0, 2x - y - 2z = 0, x - y = 0$) [4]
Oct/Nov 2011/13/11O

35. Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{pmatrix} 4 & -1 & 1 \\ -1 & 0 & -3 \\ 1 & -3 & 0 \end{pmatrix}$.
- $$\left(\lambda = -3, 2, 5; \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right)$$

Find a non-singular matrix P and a diagonal matrix D such that $A^5 = PDP^{-1}$.

$$\left(P = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}, D = \begin{pmatrix} -243 & 0 & 0 \\ 0 & 32 & 0 \\ 0 & 0 & 3125 \end{pmatrix} \right)$$

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36. A 3×3 matrix A has eigenvalues $-1, 1, 2$, with corresponding eigenvectors

$$\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

respectively. Find

- (i) the matrix A , $\left(A = \begin{pmatrix} 1.5 & 0.5 & 0.5 \\ 1.5 & 0.5 & 1.5 \\ -1 & 1 & 0 \end{pmatrix} \right)$
- (ii) A^{2n} , where n is a positive integer. $\left(A^{2n} = \frac{1}{2} \begin{pmatrix} 2^{2n} + 1 & 2^{2n} - 1 & 2^{2n} - 1 \\ 2^{2n} - 1 & 2^{2n} + 1 & 2^{2n} - 1 \\ 0 & 0 & 2 \end{pmatrix} \right)$ [14]
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37. Find the eigenvalues and corresponding eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}.$$

$$\left(\lambda = 1, 3, 4; \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right)$$

Find a non-singular matrix M and a diagonal matrix D such that $(A - 2I)^3 = MDM^{-1}$, where

I is the 3×3 identity matrix.

$$\left(M = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & -1 \\ 1 & -1 & 1 \end{pmatrix}, D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 8 \end{pmatrix} \right)$$

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38. The matrix A is given by

$$A = \begin{pmatrix} 4 & 1 & -1 \\ -4 & -1 & 4 \\ 0 & -1 & 5 \end{pmatrix}.$$

Given that one eigenvector of A is $\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$, find the corresponding eigenvalue. ($\lambda = 3$) [2]

Given also that another eigenvalue of A is 4, find a corresponding eigenvector. $\begin{pmatrix} 1 \\ -4 \\ -4 \end{pmatrix}$ [2]

Given further that $\begin{pmatrix} 1 \\ -4 \\ -1 \end{pmatrix}$ is an eigenvector of A , with corresponding eigenvalue 1, find matrices P and Q , together with a diagonal matrix D , such that $A^5 = PDQ$. [6]

$$\left(P = \begin{pmatrix} 1 & 1 & 1 \\ -4 & -2 & -4 \\ 1 & -1 & -4 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 243 & 0 \\ 0 & 0 & 1024 \end{pmatrix}, Q = \begin{pmatrix} -2/3 & -1/2 & 1/3 \\ 2 & 1/2 & 0 \\ -1/3 & 0 & -1/3 \end{pmatrix} \right)$$

May/June 2010/11/12/8

39. Given that 5 is an eigenvalue of the matrix

$$A = \begin{pmatrix} 5 & -3 & 0 \\ 1 & 2 & 1 \\ -1 & 3 & 4 \end{pmatrix},$$

find a corresponding eigenvector. $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ [2]

Hence find an eigenvalue and a corresponding eigenvector for the matrix $A + A^2$. ($\lambda =$ [2]

$$30, \begin{pmatrix} 1 \\ -4 \\ -4 \end{pmatrix})$$

May/June 2010/13/1

40. The matrix

$$A = \begin{pmatrix} 3 & 1 & 4 \\ 1 & 5 & -1 \\ 2 & 1 & 5 \end{pmatrix}$$

has eigenvalues 1, 5, 7. Find a set of corresponding eigenvectors. $\left(\begin{pmatrix} 17 \\ -6 \\ -7 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)$ [5]

Find a matrix P and a diagonal matrix D such that $A^n = PDP^{-1}$.
[The evaluation of P^{-1} is not required.] [3]

$$\left(P = \begin{pmatrix} 17 & 1 & 1 \\ -6 & -2 & 0 \\ -7 & 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5^n & 0 \\ 0 & 0 & 7^n \end{pmatrix} \right)$$

Determine the set of values of the real constant k such that $k^n A^n$ tends to the zero matrix as $n \rightarrow \infty$. $\left(-\frac{1}{7} \leq k \leq \frac{1}{7} \right)$ [3]

May/June 2009/1/9

41. The matrix A has λ as an eigenvalue with e as a corresponding eigenvector. Show that e is an eigenvector of A^2 and state the corresponding eigenvalue. (λ^2) [2]

Given that one eigenvalue of A is 3, find an eigenvalue of the matrix $A^4 + 3A^2 + 2I$, justifying your answer. **(110)** [3]

Oct/Nov 2008/1/4

42. Show that if λ is an eigenvalue of the square matrix A with e as a corresponding eigenvector, and μ is an eigenvalue of the square matrix B for which e is also a corresponding eigenvector, then $\lambda + \mu$ is an eigenvalue of the matrix $A + B$ with e as a corresponding eigenvector. [2]

The matrix

$$A = \begin{pmatrix} 3 & -1 & 0 \\ -4 & -6 & -6 \\ 5 & 11 & 10 \end{pmatrix}$$

has $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ as an eigenvector. Find the corresponding eigenvalue. $(\lambda = 4)$ [1]

The other two eigenvalues of A are 1 and 2, with corresponding eigenvectors $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$ respectively. The matrix B has eigenvalues 2, 3, 1 with corresponding eigenvectors $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$ respectively. Find a matrix P and a diagonal matrix D such that $(A + B)^4 = PDP^{-1}$.

$$\left(P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ -2 & -3 & 1 \end{pmatrix}, D = \begin{pmatrix} 81 & 0 & 0 \\ 0 & 256 & 0 \\ 0 & 0 & 1296 \end{pmatrix} \right) \quad [3]$$

May/June 2008/1/3

43. Find the eigenvalues of the matrix

$$A = \begin{pmatrix} -1 & 1 & 4 \\ 1 & 1 & -1 \\ 2 & 1 & 1 \end{pmatrix}$$

and corresponding eigenvectors.

$$\left(\lambda = -3, 1, 3; \begin{pmatrix} -17 \\ 6 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)$$

The matrix B is defined by

$$B = A - kI,$$

where I is the 3×3 identity matrix and k is a real number. Find a non-singular matrix P and a diagonal matrix D such that

$$B^3 = PDP^{-1}.$$

$$\left(P = \begin{pmatrix} -17 & 1 & 1 \\ 6 & -2 & 0 \\ 7 & 1 & 1 \end{pmatrix}, D = \begin{pmatrix} -(3+k)^3 & 0 & 0 \\ 0 & (1-k)^3 & 0 \\ 0 & 0 & (3-k)^3 \end{pmatrix} \right)$$

Oct/Nov 2007/1/11

44. The matrix A , given by

$$A = \begin{pmatrix} 7 & -4 & 6 \\ 2 & 2 & 2 \\ -3 & 4 & -2 \end{pmatrix},$$

has eigenvalues 1, 2, 4. Find a set of corresponding eigenvectors. $\left(A: \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right)$

[4]

Hence find the eigenvalues of B , where

$$B = \begin{pmatrix} 10 & -4 & 6 \\ 2 & 5 & 2 \\ -3 & 4 & 1 \end{pmatrix},$$

and state a set of corresponding eigenvectors.

$$\left(B: 4, 5, 7; \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right) \quad [3]$$

May/June 2007/1/6

45. It is given that

$$A = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & -3 \end{pmatrix}.$$

Write down the eigenvalues of A and find the corresponding eigenvectors.

[5]

$$\left(\lambda = 1, 2, -3; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 9 \\ -4 \\ 20 \end{pmatrix} \right)$$

Oct/Nov 2006/1/1

46. The square matrix A has λ as an eigenvalue with corresponding eigenvector x . The non-singular matrix M is of the same order as A . Show that Mx is an eigenvector of the matrix B , where $B = MAM^{-1}$, and that λ is the corresponding eigenvalue.

[3]

It is now given that

$$A = \begin{pmatrix} 1 & 0 & 0 \\ a & -3 & 0 \\ b & c & -5 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(i) Write down the eigenvalues of A and obtain corresponding eigenvectors in terms of a, b, c .

[4]

$$\left(A: -5, -3, 1; \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ c \end{pmatrix}, \begin{pmatrix} 24 \\ 6a \\ ac + 4b \end{pmatrix} \right)$$

(ii) Find the eigenvalues and corresponding eigenvector of B .

[4]

$$\left(B: -5, -3, 1; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} c \\ 2 \\ c \end{pmatrix}, \begin{pmatrix} 24(1 + ac + 4b) \\ 6a \\ ac + 4b \end{pmatrix} \right)$$

(iii) Hence find a matrix Q and a diagonal matrix D such that $B^n = QDQ^{-1}$.

[3]

$$\left(Q = \begin{pmatrix} 1 & c & 24(1 + ac + 4d) \\ 0 & 2 & 6a \\ 1 & c & ac + 4b \end{pmatrix}, D = \begin{pmatrix} (-5)^n & 0 & 0 \\ 0 & (-3)^n & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)$$

May/June 2006/1/11O

47. It is given that the eigenvalues of the matrix M , where

$$M = \begin{pmatrix} 4 & 1 & -1 \\ -4 & -1 & 4 \\ 0 & -1 & 5 \end{pmatrix},$$

are 1, 3, 4. Find a set of corresponding eigenvectors. $\left(\begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ 4 \end{pmatrix} \right)$

Write down a matrix P and a diagonal matrix D such that

$$M^n = PDP^{-1},$$

where n is a positive integer.

$$\left(P = \begin{pmatrix} -1 & -1 & -1 \\ 4 & 2 & 4 \\ 1 & 1 & 4 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3^n & 0 \\ 0 & 0 & 4^n \end{pmatrix} \right)$$

Find P^{-1} and deduce that

$$\lim_{n \rightarrow \infty} 4^{-n} M^n = \begin{pmatrix} -\frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{4}{3} & 0 & \frac{4}{3} \\ \frac{4}{3} & 0 & \frac{4}{3} \end{pmatrix}.$$

$$\left(P^{-1} = \frac{1}{6} \begin{pmatrix} 4 & 3 & -2 \\ -12 & -3 & 0 \\ 2 & 0 & 2 \end{pmatrix} \right)$$

Oct/Nov 2005/1/10

48. Find the eigenvalues and a corresponding set of eigenvectors of the matrix M given by

$$M = \begin{pmatrix} a & 2 & 1 \\ 0 & b & -1 \\ 0 & 0 & c \end{pmatrix},$$

[6]

where a, b, c are all different.

$$\left(\lambda = a, b, c; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ a-b \\ 0 \end{pmatrix}, \begin{pmatrix} c-b-2 \\ a-c \\ (a-c)(b-c) \end{pmatrix} \right)$$

Find a matrix P and a diagonal matrix D such that

$$(M - kI)^n = PDP^{-1},$$

where I is the identity matrix, k is a constant scalar and n is a positive integer.

[4]

[You are not required to evaluate P^{-1} .]

$$\left(P = \begin{pmatrix} 1 & -2 & c-b-2 \\ 0 & a-b & a-c \\ 0 & 0 & (a-c)(b-c) \end{pmatrix}, D = \begin{pmatrix} (a-k)^n & 0 & 0 \\ 0 & (b-k)^n & 0 \\ 0 & 0 & (c-k)^n \end{pmatrix} \right)$$

May/June 2005/1/9

- 49.** The matrix A has λ as an eigenvalue with e as a corresponding eigenvector. Show that if A is non-singular then

(i) $\lambda \neq 0$, [2]

(ii) the matrix A^{-1} has λ^{-1} as an eigenvalue with e as a corresponding eigenvector. [2]

The matrices A and B are given by

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & -2 & 4 \\ 0 & 0 & -3 \end{pmatrix} \quad \text{and} \quad B = (A + 4I)^{-1}.$$

Find a non-singular matrix P , and a diagonal matrix D , such that $B = PDP^{-1}$. [10]

$$\left(P = \begin{pmatrix} 1 & 1 & 3 \\ 0 & 3 & 8 \\ 0 & 0 & -2 \end{pmatrix}, D = \begin{pmatrix} 1/5 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)$$

Oct/Nov 2004/1/120

- 50.** Write down the eigenvalues of the matrix A , given by

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 3 & -1 \\ 0 & 0 & 4 \end{pmatrix},$$

and obtain a set of corresponding eigenvectors.

$$\left(\lambda = 1, 3, 4; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 3 \\ -3 \end{pmatrix} \right)$$

Find a non-singular matrix P and a diagonal matrix D such that $A^5 = PDP^{-1}$.

$$\left(P = \begin{pmatrix} 1 & 1 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & -3 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3^5 & 0 \\ 0 & 0 & 4^5 \end{pmatrix} \right)$$

May/June 2004/1/5

- 51.** Find the eigenvalues and corresponding eigenvectors of the matrix A , where

$$A = \begin{pmatrix} 6 & 4 & 1 \\ -6 & -1 & 3 \\ 8 & 8 & 4 \end{pmatrix}.$$

$$\left(\lambda = -1, 2, 8; \begin{pmatrix} 4 \\ -9 \\ 8 \end{pmatrix}, \begin{pmatrix} 5 \\ -6 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right)$$

Hence find a non-singular matrix P and a diagonal matrix D such that $A + A^2 + A^3 = PDP^{-1}$.

$$\left(P = \begin{pmatrix} 4 & 5 & 1 \\ -9 & -6 & 0 \\ 8 & 4 & 2 \end{pmatrix}, D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 584 \end{pmatrix} \right)$$

Oct/Nov 2003/1/10

52. The vector e is an eigenvector of each of the $n \times n$ matrices A and B , with corresponding eigenvalues λ and μ respectively. Prove that e is an eigenvector of the matrix AB with eigenvalue $\lambda\mu$. [3]

Find the eigenvalues and corresponding eigenvectors of the matrix C , where

$$C = \begin{pmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 2 & 1 & 2 \end{pmatrix}. \quad [8]$$

$$\left(C: -2, 2, 4; \begin{pmatrix} -17 \\ 6 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)$$

Verify that one of the eigenvectors of C is an eigenvector of the matrix D , where

$$D = \begin{pmatrix} -3 & 1 & 1 \\ 0 & -2 & 4 \\ 0 & 0 & -4 \end{pmatrix}. \quad [2]$$

Hence find an eigenvalue of the matrix CD . (-8)

May/June 2003/1/110

53. The vector e is an eigenvector of the square matrix G . Show that
 (i) e is an eigenvector of $G + kI$, where k is a scalar and I is an identity matrix,
 (ii) e is an eigenvector of G^2 . [5]

Find the eigenvalues, and corresponding eigenvectors, of the matrices A and B^2 , where

$$A = \begin{pmatrix} 3 & -3 & 0 \\ 1 & 0 & 1 \\ -1 & 3 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -5 & -3 & 0 \\ 1 & -8 & 1 \\ -1 & 3 & -6 \end{pmatrix}. \quad [9]$$

$$\left(A: 0, 2, 3; \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}; B: -8, -6, -5; \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right)$$

54. Find the eigenvalues of the matrix

$$\begin{pmatrix} 4 & -2 \\ 3 & -1 \end{pmatrix}.$$

Find also corresponding eigenvectors. $\left(\lambda = 1, 2; e = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$ [5]

May/June 2002/1/1